Ellipsis Investigation in Altaic Languages

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Introduction

Lin · guis · tics

the scientific study of language and its structure.

Semantics

the study of linguistic **meaning**: what is meaning, how words get their meaning, etc.

Linguistics

- Phonetics
- Pragmatics
- Morphology
- Semantics
- Syntax

Syntax

the study of how **well-formed** sentences and phrases are created in language

Ellipsis...

Words removed **[elided]** from a sentence that are still understood with the proper context.

Someone left.
Guess who ~~[left]~~?

Verb Phrase Ellipsis

Sluicing

Argument Ellipsis

Fragments

Gapping

Let’s take a look at some examples

Verb Phrase Ellipsis (VPE)

A specific type of ellipsis involving the deletion of the entire verb phrase.

01 | *English VPE*

1) Niko cooks and **Yağmur does, too**.
   Niko cooks and Yağmur does ~~[cook]~~, too.

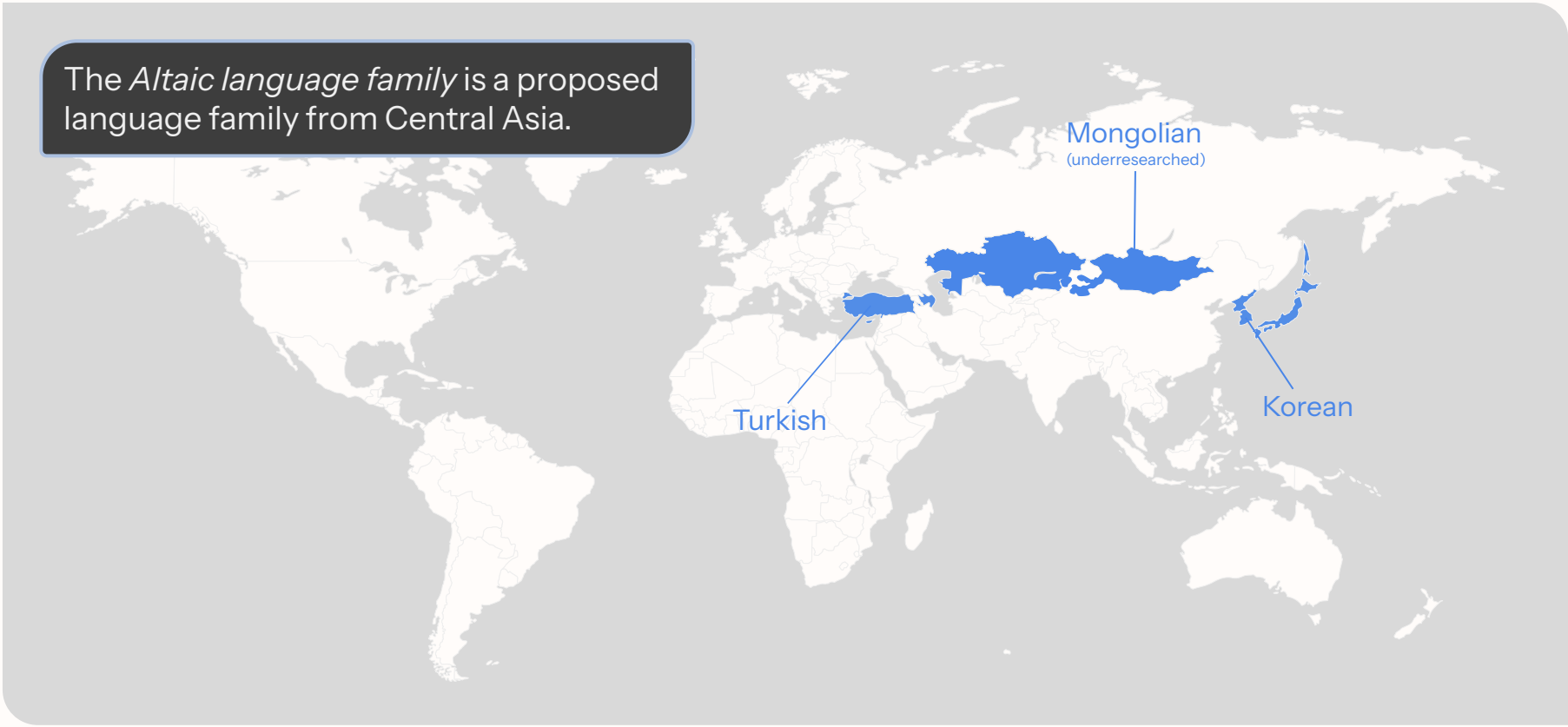
2) Niko ate an apple and **Yağmur did, too**.
   Niko ate an apple and Yağmur did ~~[eat an apple]~~, too.

Overarching Research Question

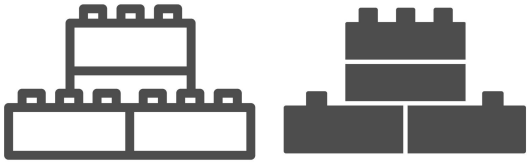
What do the VPE-like constructions in Altaic languages reveal about the cross-linguistic nature of verbal ellipsis?

“Altaic” Languages

The *Altaic language family* is a proposed language family from Central Asia.



Why Altaic Languages?



Common Structure

Same word order (SOV)
Agglutinative type
- ➔ Suffix morphemes
- ➔ “Lego bricks”



Understudied

English + Indo-European dominate linguistics research
- ➔ Little linguistic analysis in some “Altaic”languages



Contrastive Views

Differing views across literature
- ➔ Some refute existence of ellipsis while some support this construction

Focused Research Questions

01

Do Turkish and Mongolian exhibit VPE-like constructions comparable to those documented in Korean?

02

What diagnostics distinguish *genuine* VP-ellipsis from “superficially” similar constructions?

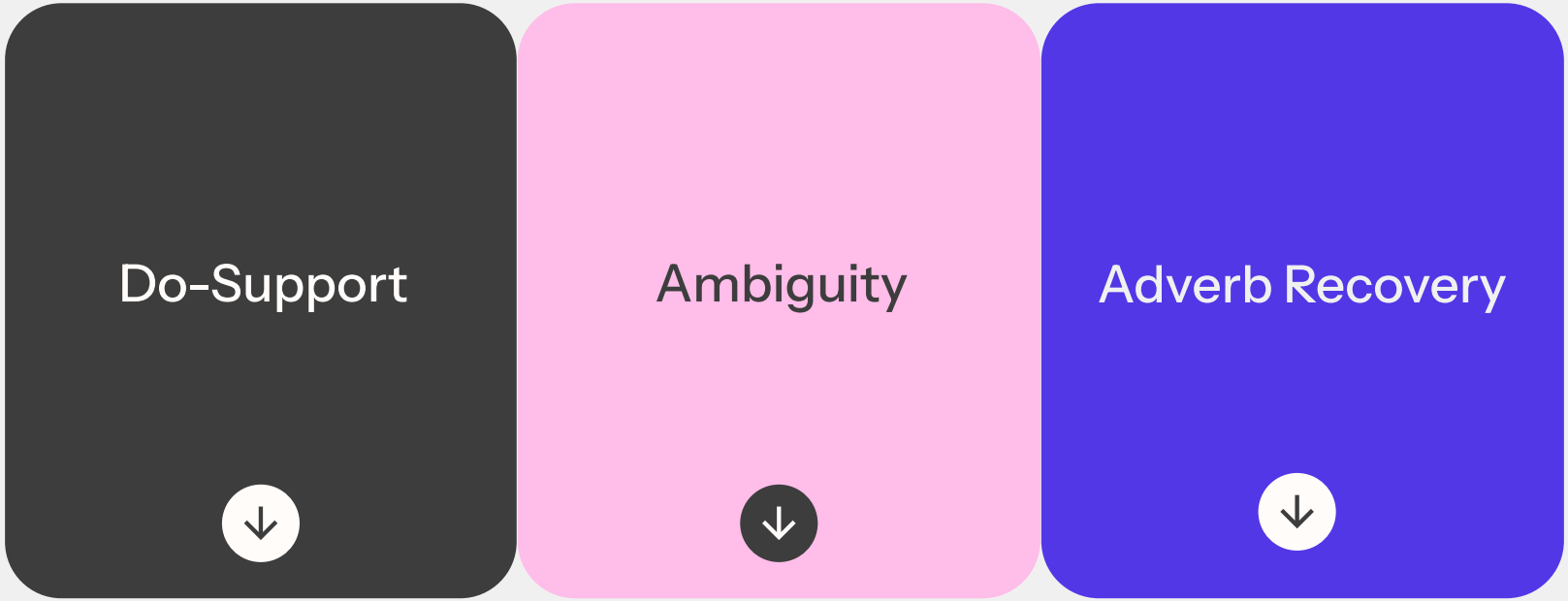
03

What do the diagnostics show about the compatibility of VPE-like constructions in Korean, Turkish and Mongolian with *genuine VPE* like in English?

04

Could the diagnostics reveal whether the target of deletion is a verbal projection (VP) or a nominal projection (NP) in these constructions?

English VPE Diagnostics



Do-Support

The auxiliary verb *do* (or its derivations) coming in to preserve information when the VP is elided.

01 | *Do-support (English VPE)*

Niko cooks and **Yağmur does** [-]**, too.**

Niko cooks and Yağmur ~~[cooks]~~ too.



“Does” preserves tense and is **required** for VPE in English.

Ambiguity

The use of VPE in English can lead to multiple ambiguous meanings: strict and sloppy readings.

01 | *Ambiguity in English VPE*

John likes his dog and **Bill does** [–]**, too.**

John likes his dog and Bill ~~[likes his dog]~~ too.



STRICT
Bill likes John's dog.

SLOPPY
Bill likes his own dog.

Adverb Recovery

If a manner adverb (which tells us how the action is performed) is part of the VP, it is recovered.

01 | *Adverb Recovery in English VPE*

John runs **fast** and Mary does [-], too.

*John runs fast and Mary [runs **fast**], too.*



As English speakers, we get the interpretation that Mary not only **runs**, but **runs fast**.

VPE-*like* Constructions



Note on Do-Support

Do-support (used in questions, negative sentences) is not observed the same way it is observed in English, i.e. as a separate word that bears tense information.

- *do* replaces the main verb in VPE in English and it bears tense
- although there are similar verbs in these languages (aka light verbs such as *make* or *do* in English), they do not replace any main verb in ellipsis

Additive

01 | *Additive VPE* (Lee 2017)

Johnun cakiuy pyencilul pelieessta. Mary**to** peliessta.
John his letter threw.away. Mary.also threw.away.
“John threw away his letter. Mary did too.”

The use of an **additive particle** like *also* or *too* to emphasize that the **previous phrase (antecedent)** also applies to the **later ellipsis site.**

➔ In this sentence, the additive particle ***-to*** “also” is used.

Additive | Ambiguity

01 | *Korean* (Lee 2017)

Johnun cakiuy pyencilul pelieessta. Mary**to** peliessta.
John self letter threw.away Mary.also threw.away.
‘John threw away his letter. Mary threw away, too.’ > his (John’s) letter or her own letter

✓

02 | *Turkish* (Own Data)

Çayan annesini özlüyor. Yağmur **da** özlüyor.
Çayan his.mother missing Yağmur too missing
‘Çayan misses his mother. Yağmur misses too.’ > his (Çayan’s) mother or her own mother

✓

03 | *Mongolian* (Own Data)

Enja eejiigee shüümjilsen. Bold ch gesen shüümjilsen.
Enja her.mother criticized Bold as well criticized
‘Enja criticized her mother. Bold criticized too.’

?

Additive | Adverb Recovery

01 | *Korean* (Hildebrandt 2021)

M.nun tangkunul ppalli thongccaylo samkhyessko
Minsu carrot quickly whole devour
'Minsu quickly devoured the carrot whole,

Swuni**to** samkhyesseyo.
S-also devour
and Suni also devoured.'

?

02 | *Turkish* (Own Data)

Çayan havucu çabucak yedi
Çayan carrot quickly ate
'Çayan ate the carrot quickly

ve Yağmur da yedi.
and Yağmur too ate
and Yağmur ate too.' ~~**(quickly)**~~

×

03 | *Mongolian* (Own Data)

Enja ter nomyg khurdan unshsan,
Enja that book quickly read
Enja read that book quickly,

bas Nomin ch gesen unshsan.
too Nomin too read
and Nomin read too.

?

Contrastive

01 | *Contrastive VPE* (Lee 2017)

Johnun [cakiuy kay]wa kotcal sanpohanta.
John self dog quite.often takes a walk
'John quite often takes a walk with his dog.'

Kulena Billun kotcal ttaylinta.
But Bill quite.often beats
'But Bill quite often beats his dog.'

The use of either a **verb mismatch** or a **verb negation** to form a contrastive focus, where one verb phrase is **different or the opposite** of another.

➔ Here, there is a verb mismatch with *takes a walk* and *beats*.

Contrastive | Ambiguity

01 | *Korean* (Lee 2017)

John-un [caki-uy kay]-wa kotcal sanpoha-n-ta.
J-Top self-Gen dog quite.often take.a.walk
'John quite often takes a walk with his dog.'

Kulena Bill-un [e] kotcal sanpoha-ci anh-nun-ta.
But Bill-Top quite.often take.a.walk not.do
'But Bill does not quite often take a walk with his dog.'
> John's dog or Bill's dog

02 | *Turkish* (Own Data)

Çayan annesini özlüyor
Çayan mother miss
'Çayan misses his mother,

ama Yağmur özlemiyor
but Yağmur miss not
but Yağmur doesn't. > his mother or her mother

03 | *Mongolian* (Sakamoto 2017)

Bat-Ø [CP uuri-n bagsh-Ø awiyastai gej] bodo-j bai-gaa
Bat-NOM self teacher genius that think be
'Bat thinks that self's teacher is a genius'

Oyuna-Ø eCP bodo-h-gui bai-gaa.
Oyuna-NOM think-INF-NEG be
'but Oyuna does not think' > Bat's teacher is a genius or Oyuna's teacher is a genius.

Contrastive | Adverb Recovery

01 | *Korean* (Hildebrandt 2021)

M.-**nun** tangkunul ppalli thongccaylo samkhyessnuntey
Minsu carrot quickly whole devour
‘Minsu quickly swallowed the carrot whole,

Swuni**nun** samkhicianhassta
Swuni devoured
but Suni did not swallow.’

02 | *Turkish* (Şener and Takahashi 2010)

Can sorunu hızla çözdü.
Can problem quickly solved
‘John solved the problem quickly.

Filiz-se çözmedi.
Phylis-however didn’t.solve
Phylis did not solve the problem (~~quickly~~).’

03 | *Mongolian* (Sakamoto and Bao 2020)

Bayatur-bol qurduniyar üjümun ariqiyi uuyujai.
Bagatur quickly grape alcohol drank
‘Bagatur drank wine quickly.

U.-bol almuradun ariqiyi uuyuysan ügei.
Ulagan apple alcohol drank not
Ulagan did not drink wine (~~quickly~~).’

Light Verb VPE

01 | *Light Verb VPE* (Own Data)

Yenghuyka chwulpal-**ha**yssta,	Chelsooto kuli-**ha**yssta
Yenghuy begin-did	Chelsoo.also that-did
‘Yenghi left,	**Chelswu also did that.’**

A **light verb** is a verb with very **little tangible meaning** on its own, but forms a complex predicate when it comes together with **another element.**

➔ Here, the light verb is *ha*, which is meaningless on its own.

Light Verb | Ambiguity

01 | *Korean* (Own Data)

Yenghuyka caki emmahako chwulpal-hayssta.
Yenghuy self mother.with begin-did
'Yenghi left with her mother,

Chelsooto kuli-hayssta
Chelsoo.also that-did
Chelswu also did that.'

02 | *Turkish* (Own Data)

Çayan arkadaşını ikna etti,
Çayan his.friend persuasion did
'Çayan convinced his friend,

Yağmur da etti.
Yağmur too did
Yağmur did too.' > **his (Çayan's) friend or her own friend**

03 | *Mongolian* (Own Data)

Enja övchtei bolson.
Enja sick became.
'Enja became sick,

Bold ch gesen bolson.
Bold as well became
and Bold did too.'

Light Verb | Adverb Recovery

01 | *Korean* (Own Data)

Yenghuyka ppalukey chwulpal-hayssta.
Yenghuy quickly begin-did
'Yenghi left quickly,

Chelsooto kuli-hayssta
Chelsoo.also that-did
Chelswu also did that.'

✓

02 | *Turkish* (Own Data)

Çayan onu çabucak ikna etti,
Çayan him/her quickly persuasion did
'Çayan convinced him/her quickly,

Yağmur da etti.
Yağmur too did
Yağmur did too.'

✓

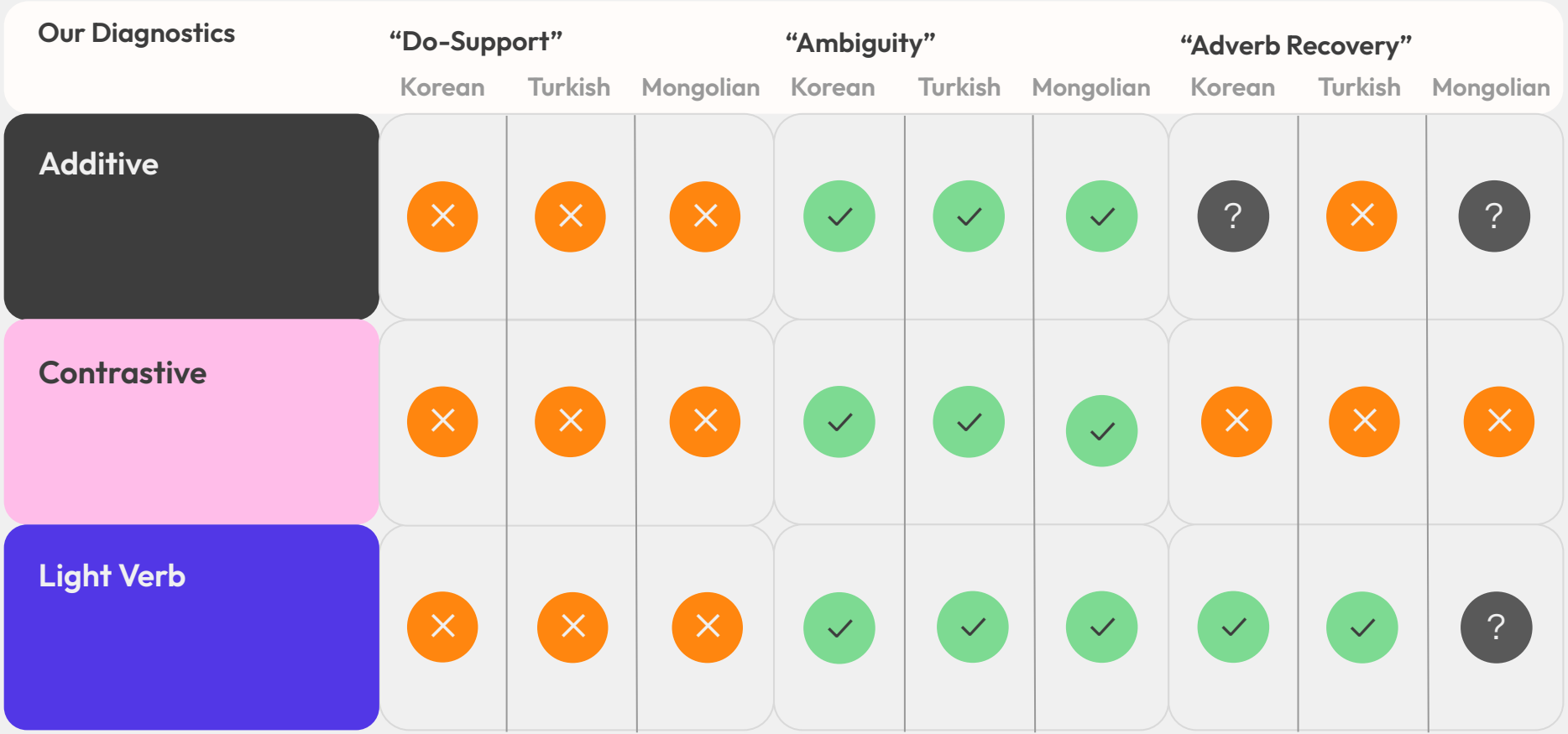
03 | *Mongolian* (Own Data)

Enja mash ikh övchtei bolson.
Enja a lot sick became.
'Enja became sick a lot,

Bold ch gesen bolson.
Bold as well became
Bold did too.'

?

Results Summary



Conclusion

Conclusion

- Analyzed common VP Ellipsis structures in 3 Altaic languages to determine cross-linguistic nature of VP Ellipsis compared to English

- Two competing hypotheses
  - Hypothesis #1: These languages exhibit a VP-ellipsis via verb movement
  - Hypothesis #2: The ellipsis applies to the noun phrase, thus these are NP ellipsis and not a VP ellipsis

- Future Work:
  - Formally test our hypothesis using existing linguistic theories
  - Answer some of these questions:
    - If the structures are the same, then why does one (proposed to) have VP ellipsis vs other not having it?
    - Why do languages that do not allow VP ellipsis allow tense to be conveyed with pseudo-sluicing?

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Thank you for making this such a memorable summer!

Q&A

Please ask us questions!

Sluicing

A construction where a question consisting only of a wh-phrase receives the full meaning conveyed in a preceding sentence (Ross, 1969).

01 | Sluicing

Someone called, but I don't know **who.**

Someone called, but I don't who ~~[called]~~.

Sluicing is introduced by a **wh-** expression.

who | what | why | when | where | how | which

Case-matching requirement:

The wh-remnant in the elliptical clause must bear *the same case* as its correlate in the full clause.

Er will	jemandem	schmeicheln,	aber	sie	wissen
he wants	someone.**DAT**	flatter	but	they	know

nicht, ***wen**/**wem.**
not **who.ACC**/**who.DAT**

'He wants to flatter someone, but they don't know who.'

The **correlate** of a wh-phrase is what the phrase is referring to in the **antecedent** (the sentence before). The correlate of *who* in 01 is *someone*.

Sluicing in Korean

Sluicing does exist in Korean, but it has **numerous unique features**;

- ***Wh*-in-situ**: *Wh*-phrases stay in their base position, even though most languages have overt *wh*-movement.
- **Obligatory/Common presence of COP verb**: The copular verb *i-ta* is often needed to make constructions grammatical.
- **Case-Matching Requirements**: The response must case-match the question, either in bare or Accusative case.
- **Predicate Embedding Requirement**: Embedded sluices must be a predicate with the interrogative marker *-nci*, not just a *wh*-phrase.
- **Optionality of Postpositions**: A fragment sluice doesn't need a postposition, which is a modifier that comes after it's headword.
- **Multiple *wh*-remnants**: Sluicing constructions can have multiple wh-remnants.
- **Island Insensitivity**: The correlate of the *wh*-expression can be located inside an island in the antecedent.
- **Subject Pronoun *ku kes***: Constructions can include the subject pronoun *ku kes* to refer to the event in the antecedent.
- **Use of QUE value instead of WH**: Sluicing relies on the QUE (question) value from *-nci*, instead of the WH value like English.

Because the previous Pseudocleft hypothesis **couldn't explain all of these features accurately**, researchers like Kim, Cho (2013) believe that **sluices are generated "as-is,"** without movement or deletion.

Sluicing in Mongolian (SLCs)

01 | Case Matching Violations

Wh-remnants always take the accusative case.

Hen_negen-**Ø** ene nom-ig zeelle-sen.
someone-**NOM** this book-ACC borrow-PERF
‘Someone borrowed this book.’

→

Gevch, bi hen***-Ø**/**-ig** ni med-eh-gui.
but I who-**NOM**/**ACC** NI know-INF-NEG
‘But, I don’t know who.’

02 | Non-Wh Remnants

Unlike in English, Mongolian allows sluicing with non-wh phrases (nouns, adj, etc.) as remnants.

Bat-Ø Ceren-ig uchigdur Sendai-d uhri-n hel-ig id-sen gej bodo-j bai-gaa.
B-NOM C-ACC yesterday Sendai-LOC cow-GEN tongue-ACC eat-PERF that think-INF be-NPST.
‘(lit.) Bat thinks that Ceren ate cow tongue in Sendai yesterday.’

Gevch, bi **sushi** gej bodo-j bai-gaa.
but I sushi that think-INF be-NPST.
‘(lit.) But, I think that sushi.’

‘I think that Ceren ate sushi in Sendai yesterday.’

Gevch, bi **unudru** gej bodo-j bai-gaa.
but I today that think-INF be-NPST.
‘(lit.) But, I think that today.’

‘I think that Ceren ate cow tongue in Sendai today.’

Sluicing in Mongolian (SLCs)

01 | Multiple Remnants

Unlike English, Mongolian SLCs can have multiple remnants.

Bat-Ø hen_negen-d hen_negen_zuil-ig ug-sun.
B-NOM someone-DAT something-ACC give-PERF
'Bat gave something to someone.'

Gevch, bi hen-d yu-g ni med-eh-gui.
but I who-DAT what-ACC NI know-INF-NEG
'But I don't know to whom what.'

02 | Matrix

Unlike in English, Mongolian allows sluicing with non-wh phrases (nouns, adj, etc.) as remnants.

Bat-Ø Ceren-ig uchigdur Sendai-d uhri-n hel-ig id-sen gej bodo-j bai-gaa.
B-NOM C-ACC yesterday Sendai-LOC cow-GEN tongue-ACC eat-PERF that think-INF be-NPST.
'(lit.) Bat thinks that Ceren ate cow tongue in Sendai yesterday.'

Gevch, bi **sushi** gej bodo-j bai-gaa.
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'(lit.) But, I think that sushi.'

'I think that Ceren ate sushi in Sendai yesterday.'

Gevch, bi **unudru** gej bodo-j bai-gaa.
but I today that think-INF be-NPST.
'(lit.) But, I think that today.'

'I think that Ceren ate cow tongue in Sendai today.'

Korean VPE

A. Anna-nun [caki-ka khu-ta-ko sayngkak] an-hay.
Anna-TOPIC self-NOMINATIVE tall-DECLARATIVE-C thought NEGATIVE-HA.DECLARATIVE
‘Anna doesn’t think that she is tall.’

B. Mary-nun [*e*] hay.
Mary-TOPIC.CONTRASTIVE HA.DECLARATIVE
‘Mary does, though.’

The VN-*ha* Construction: A structure that allows for VPE in Korean, consisting of a **VN** + a meaningless morpheme “*ha*.” (Park 2025)

VN (Verbal Noun): a noun (person, place, or thing) that is acting as or derived from a verb (action word).

Korean VPE cont.

Two VPE Types (Lee 2017)

Chelswu-ka cha-lul sa-ass-ta.
Chelswu-NOM car-ACC buy-PST-DECL
'Chelswu bought a car.'

A) Additive

Yengswu-**to** [e] sa-ass-ta
Yengswu-also [e] buy-PST-DECL
'Yengswu bought a car, also.'

B) Contrastive

Yengswu-**nun** [e] an.sa-ass-ta.
Yengswu-CONT [e] not.buy-PST-DECL
'Yengswu did not buy a car.'

Adverbial Argument

Goldberg claims that manner/purpose adverbials cannot be elided → no genuine VPE

John-i ppali tali-ko Mary-to *(ppali)
John-NOM fast run-CONJ Mary-also *(fast)
tali-nta.
run-DECL

✕ ungrammatical without *ppali* 'fast'

'John runs fast and Mary runs fast too.'

However, this fails to capture the general pattern.

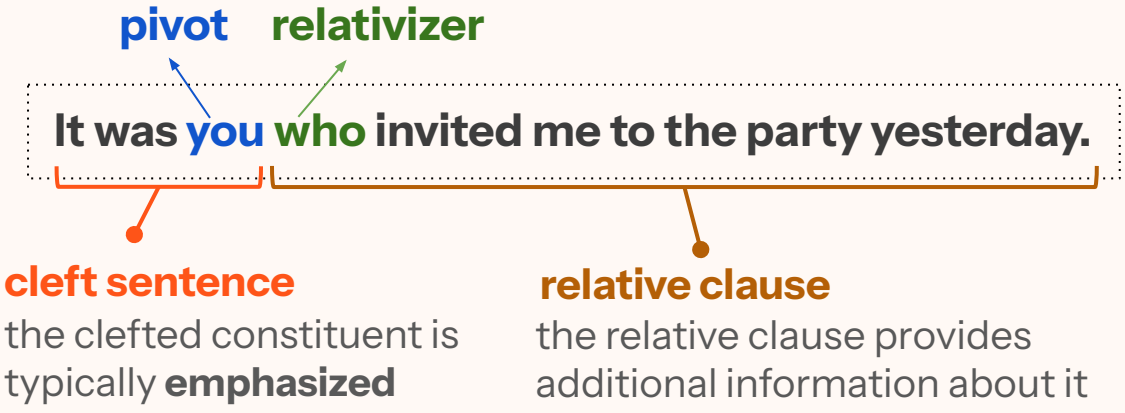
Swuni-ka panci-lul kum-ulo ppalli yeyppu-key mantul-ess-e-yo?
Sw-NOM ring-ACC gold-INSTR quickly pretty-RES make-PST-DECL-POL?
'Did Suni quickly make a ring out of gold pretty?'

Yey, mantul-ess-e-yo.
Yes, make-PST-DECL-POL
Yes, made. ⟹

✓ quickly
✓ out of gold
✓ pretty

Cleft

Cleft sentences often consist of a main clause with a copula and a relative clause – a subordinate clause introduced by a relativizer.



Cleft - Mongolian

1 The pivot must bear nominative case

Bat-∅ Oyuna-d nom-ig ug-sun
Bat-NOM Oyuna-DAT **book-ACC** give-PERF
'Bat gave the book to Oyuna.'

Bat-in Oyuna-d ug-sun ni nom-∅/*-ig
B-GEN O-DAT give-PERF NI **book-NOM**/ACC
'It is the book that Bat gave to Oyuna.'

- **The pivot nom (book) must be in nominative case in cleft sentence**

2 The pivot can be wh-phrases

Ta_nar-in hamgin (undru) ni hen-∅ be?
you.PL-GEN most tall NI **who-NOM** Q
'Who is it that is the tallest of you?'

The pivot is the wh-phrase *hen* (who).

- **The sentence with pivot wh-phrase is grammatical**

Cleft - Mongolian

3 The copulas can follow the pivot

Oyuna-gin id-sen ni tort-∅ **bai-na**
Oyuna-GEN eat-PERF NI cake-NOM **be-PRES**
It is the cake that Oyuna ate.

Oyuna-gin id-sen ni tort-∅ **bai-san**
Oyuna-GEN eat-PERF NI cake-NOM **be-PERF**
It was the cake that Oyuna ate.

- **The copulas can optionally follow the pivot tort (cake)**
- **Assumption: the insertion of copulas contributes to the emphasis of tense**

4 The adjuncts cannot be in the pivot

*Oyuna-gin ene nom-ig hudaldag-av-san ni **uchigdur**

Oyuna-GEN this book-ACC buy-take-PERF NI **yesterday**
It was yesterday that Oyuna bought this book.

- **The adjunct, uchigdur (yesterday) is incompatible with the cleft sentence**
- **The sentence is ungrammatical**

Cleft - Mongolian

5 More than one element in the pivot is ungrammatical

*Bat-∅ ug-sun ni **Oyuna(-d)** **nom(-ig)**.
Bat-NOM give-PERF NI **Oyuna-DAT** **book-ACC**
(lit.) It is to Oyuna books that Bat gave.

Two elements in the pivot: Oyuna & *nom* (book)

*Ug-sun ni **Bat-∅** **Oyuna(-d)** **nom(-ig)**.
give-PERF NI **Bat-NOM** **Oyuna-DAT** **book-ACC**
(lit.) It is Bat to Oyuna books that gave.

Three elements: Bat, Oyuna, and nom (book)